SULFURIC ACID, concentrated (> 51% and < 100%) **ICSC: 0362 (November 2016)**
Sulphuric acid
Oil of vitriol

CAS #: 7664-93-9
UN #: 1830
EC Number: 231-639-5

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Not combustible. Many reactions may cause fire or explosion. Gives off irritating or toxic fumes (or gases) in a fire. Risk of fire and explosion on contact with bases, combustible substances, reducing agents, water or organic materials.	NO contact with incompatible materials: See Chemical Dangers	NO water. In case of fire in the surroundings, use appropriate extinguishing media. In case of fire: keep drums, etc., cool by spraying with water. NO direct contact of the substance with water.

PREVENT GENERATION OF MISTS! AVOID ALL CONTACT! IN ALL CASES CONSULT A DOCTOR!			
	SYMPTOMS	**PREVENTION**	**FIRST AID**
Inhalation	Cough. Sore throat. Burning sensation. Shortness of breath. Laboured breathing.	Use ventilation, local exhaust or breathing protection.	Fresh air, rest. Half-upright position. Artificial respiration may be needed. Refer immediately for medical attention.
Skin	Redness. Pain. Blisters. Serious skin burns.	Protective gloves. Protective clothing. Apron.	Wear protective gloves when administering first aid. First rinse with plenty of water for at least 15 minutes, then remove contaminated clothes and rinse again. Refer immediately for medical attention.
Eyes	Redness. Pain. Severe burns.	Wear face shield or eye protection in combination with breathing protection.	Rinse with plenty of water for several minutes (remove contact lenses if easily possible). Refer immediately for medical attention.
Ingestion	Burns in mouth and throat. Burning sensation behind the breastbone. Abdominal pain. Vomiting. Shock or collapse.	Do not eat, drink, or smoke during work.	Rinse mouth. Give nothing to drink. Do NOT induce vomiting. Refer immediately for medical attention.

SPILLAGE DISPOSAL

Evacuate danger area! Consult an expert! Personal protection: chemical protection suit including self-contained breathing apparatus. Do NOT let this chemical enter the environment. Do NOT absorb in saw-dust or other combustible absorbents. Collect leaking liquid in sealable containers. Absorb remaining liquid in dry sand or inert absorbent. Then store and dispose of according to local regulations. Cautiously neutralize remainder with lime or soda ash.

STORAGE

Dry. Separated from food and feedstuffs and incompatible materials. See Chemical Dangers. Store only in original packaging.

PACKAGING

Unbreakable packaging.
Put breakable packaging into closed unbreakable container.
Do not transport with food and feedstuffs.

CLASSIFICATION & LABELLING

According to UN GHS Criteria

DANGER

Fatal if inhaled
Causes severe skin burns and eye damage
May cause respiratory irritation
May be corrosive to metals
See Notes

Transportation
UN Classification
UN Hazard Class: 8; UN Pack Group: II

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PHYSICAL & CHEMICAL INFORMATION

Physical State; Appearance
ODOURLESS COLOURLESS OILY HYGROSCOPIC LIQUID.

Physical dangers
No data.

Chemical dangers
Decomposes on heating. This produces toxic and corrosive gases including sulfur oxides. The substance is a strong oxidant. It reacts with combustible and reducing materials and organic materials. This generates fire and explosion hazard. The substance is a strong acid. It reacts violently with bases and is corrosive to most common metals forming a flammable/explosive gas (hydrogen - see ICSC 0001). Reacts violently with water. This generates heat and fire or explosion hazard. See Notes. Attacks many plastics.

Formula: H_2SO_4
Molecular mass: 98.1
Decomposes at 340°C
Melting point: 10°C
Relative density (water = 1): 1.8 (20°C)
Solubility in water at 20°C: miscible
Vapour pressure, Pa at 20°C: < 10 (negligible)
Relative vapour density (air = 1): 3.4

EXPOSURE & HEALTH EFFECTS

Routes of exposure
Serious local effects by all routes of exposure. The substance can be absorbed into the body by inhalation of its aerosol.

Effects of short-term exposure
The substance is very corrosive to the eyes, skin and respiratory tract. Corrosive on ingestion. Exposure could cause asphyxiation due to swelling in the throat. Inhalation of high concentrations may cause lung oedema, but only after initial corrosive effects on the eyes and the upper respiratory tract have become manifest. Inhalation may cause asthma-like reactions (RADS). Medical observation is indicated. See Notes.

Inhalation risk
Evaporation at 20°C is negligible; a harmful concentration of airborne particles can, however, be reached quickly when dispersed.

Effects of long-term or repeated exposure
Repeated or prolonged contact with skin may cause dermatitis. Repeated or prolonged inhalation may cause effects on the lungs. Risk of tooth erosion upon repeated or prolongated exposure to an aerosol of this substance. Mists of this strong inorganic acid are carcinogenic to humans. See Notes.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 0.2 mg/m^3, as TWA; A2 (suspected human carcinogen).
MAK: (inhalable fraction): 0.1 mg/m^3; peak limitation category: I(1); carcinogen category: 4; pregnancy risk group: C.
EU-OEL: 0.05 mg/m^3 as TWA

ENVIRONMENT

The substance is harmful to aquatic organisms.

NOTES

The symptoms of lung oedema often do not become manifest until a few hours have passed and they are aggravated by physical effort. Rest and medical observation are therefore essential.
IARC considers mists of strong inorganic acid to be carcinogenic (group 1). However there is no information available on the carcinogenicity of other physical forms of this substance. Therefore no classification for carcinogenicity under GHS has been applied.
NEVER pour water into this substance; when dissolving or diluting always add it slowly to the water.
Rinse contaminated clothing with plenty of water because of fire hazard.
Other UN numbers: UN1831 Sulfuric acid, fuming, hazard class 8, subsidiary hazard 6.1, pack group I; UN1832 Sulfuric acid, spent, Hazard class 8, Pack group II.

ADDITIONAL INFORMATION

EC Classification
Symbol: C; R: 35; S: (1/2)-26-30-45; Note: B